8 A particle of mass m and charge q, is initially at rest at the midpoint between two parallel plates.

A potential difference V_0 is applied across the plates.

The particle accelerates towards the upper plate and enters into a region with a uniform magnetic flux density B, directed perpendicularly out of the page. There is no magnetic field anywhere else. A snippet of the particle's path is shown in Fig. 8.1.

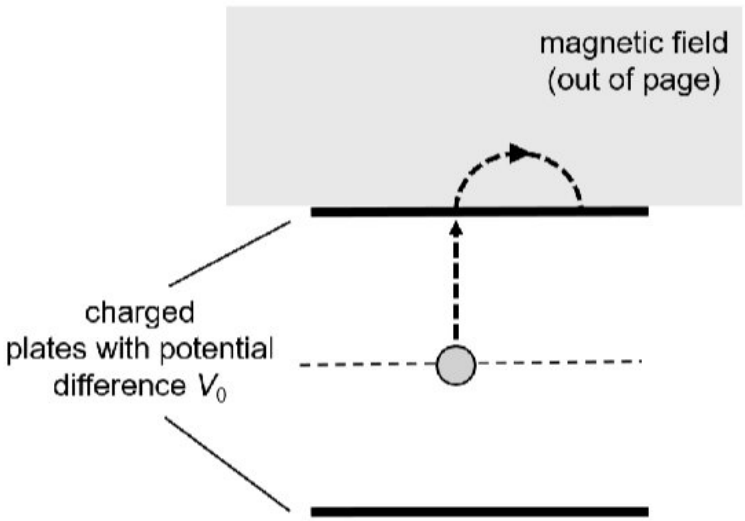


Fig. 8.1

(a) Explain

1. why the charged particle in Fig. 8.1 is positively charged,

.....

.....

.....

..... [2]

2. and which plate has a higher electric potential.

.....

.....

.....

..... [2]

(b) **(i)** Show that the magnitude of the potential difference between the top plate and the midpoint between the plates is:

$$\Delta V = \frac{V_0}{2}$$

[1]

(ii) Using your answer in **(b)(i)**, show that the velocity of the particle when it first enters the magnetic field is:

$$v = \sqrt{\frac{qV_0}{m}}$$

[1]

(c) Explain whether the speed of the particle increases while it is in the magnetic field.

.....

.....

..... [2]

(d) Show that the time travelled in the magnetic field can be expressed as

$$t = \frac{m\pi}{Bq}$$

[4]

(e) The mechanism shown in Fig. 8.1 forms the basis of a classical cyclotron.

Instead of two parallel plates, the cyclotron consists of two hollow, D-shaped metal electrodes (known as dees) separated by a narrow central gap. The dees are housed in a vacuum chamber within the uniform magnetic field of flux density B directed perpendicularly out of the page.

An alternating high-voltage source produces an electric field across the gap.

When the charged particle is released at the centre, it is accelerated across the gap into the upper dee. Shielded from the electric field inside the hollow metal dee, the magnetic field bends the particle into a semicircular path.

Upon exiting back into the gap, the electric field reverses polarity, accelerating the particle into the lower dee at a higher speed. This process repeats, producing an expanding spiral path until the particle exits the cyclotron.

This is shown in Fig. 8.2.

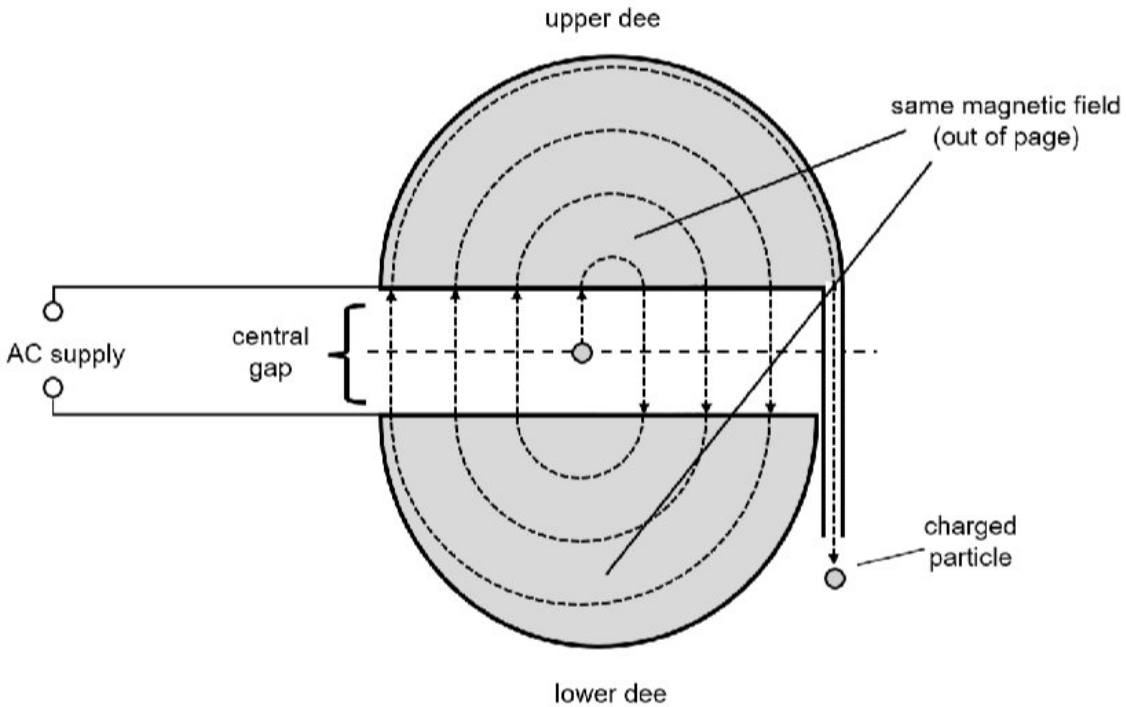


Fig. 8.2

A man claims that the time travelled in each dee is the same as that found in **(d)**.

State and explain if his claim is true.

.....

.....

.....

..... [2]

(f) The alternating voltage supply is a square wave that switches its polarity at constant time intervals equal to the time taken for the particle to complete a semicircle inside each dee.

This is shown in Fig. 8.3.

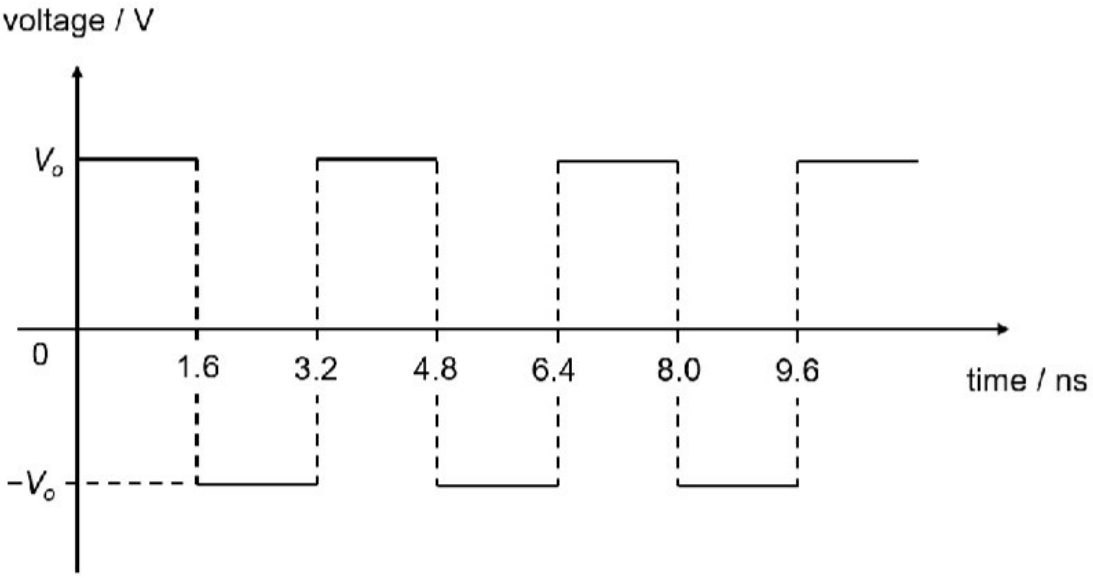


Fig. 8.3

Given that the charge to mass ratio of the particle is 1.0×10^8 C kg^{-1}, using the equation in **(d)**, determine the magnetic flux density in the dees.

You may assume that the time taken for the charged particle to travel the central gap is negligible.

magnetic flux density = ... T [2]

(g) Determine the kinetic energy of the charge when it first enters the lower dee, leaving your answer in q and V_0.

kinetic energy = ... [2]

(h) The particle leaves the cyclotron with a kinetic energy of 6.5 qV_0.

Determine the number of times it travels in the lower dee before leaving the cyclotron.

number = ... [2]

[Total: 20]